From-scratch replication: *Altermagnetism — an unconventional spin-ordered phase of matter*

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Replication report — REPLICATE-PROJECT / TEXTURES-100

July 20, 2026

Abstract

This paper is a **Perspective/review**: it states no single quantitative benchmark. Its *one testable headline claim* (abstract, Fig. 1b) is that altermagnetism combines **vanishing net magnetization** with **well-separated, conserved spin-up/spin-down channels** through an anisotropic *d-, g-, or i-wave* spin ordering, producing a **momentum-dependent spin splitting** with a symmetry-protected sign structure. We build a minimal from-scratch two-sublattice square-lattice tight-binding model of a $d_{x^2-y^2}$ altermagnet and verify all four qualitative-but-symmetry-exact ingredients.

Result: $M_{\rm net} = 0$ (exact), max spin splitting $= 0.561\, t_{nn}$, diagonal nodes to 10^{-16}, C4 antisymmetry to 10^{-16}, d-wave sign structure match = 100%. **Self-verdict: REPLICATED (mechanism, symmetry-exact).**

1 The paper's claim (the comparison anchor)

A Perspective has no fitted number. The physical content we anchor to is Fig. 1b and the abstract: in a collinear *compensated* magnet, an anisotropic higher-partial-wave (here d-wave) spin ordering yields momentum-space spin-up/spin-down iso-surfaces that are related by a **combined spin×real-space (C4) rotation**, not by translation/inversion. Consequences: (i) net magnetization is zero; (ii) the two spin channels are nonetheless split and conserved (S_z good in the non-relativistic limit); (iii) the splitting changes sign under C4 and has d-wave nodes on the Brillouin-zone diagonals.

2 Model (from scratch, numpy)

Square lattice, two magnetic sublattices A, B with opposite collinear moments $\pm m\,\hat{z}$ (Néel-compensated $\Rightarrow M_{\rm net} = 0$ by construction). A and B are related by C4, which makes their next-nearest-neighbour hopping *anisotropic and swapped*:

$$H_\sigma(k) = \begin{pmatrix} \varepsilon_A(k) + \sigma m & f(k) \\ f^*(k) & \varepsilon_B(k) - \sigma m \end{pmatrix}, \quad \sigma = \pm 1, \tag{1}$$

$$\varepsilon_A(k) = -2t_1 \cos k_x - 2t_2 \cos k_y, \qquad \varepsilon_B(k) = -2t_2 \cos k_x - 2t_1 \cos k_y, \tag{2}$$

$$f(k) = -2t_{nn}\,[\cos(k_x/2) + \cos(k_y/2)]. \tag{3}$$

No spin-orbit coupling $\Rightarrow S_z$ conserved $\Rightarrow H$ block-diagonalizes into 2×2 spin blocks. The d-wave altermagnetic form factor emerges analytically:

$$\delta(k) = \tfrac{1}{2}[\varepsilon_A(k) - \varepsilon_B(k)] = (t_1 - t_2)(\cos k_x - \cos k_y), \tag{4}$$

which is odd under $k_x \leftrightarrow k_y$ (C4) and vanishes on $k_x = \pm k_y$ (diagonal nodes) — the $d_{x^2-y^2}$ symmetry. Parameters (units of t_{nn}): $t_{nn} = 1$, $t_1 = 0.5$, $t_2 = 0.1$, $m = 0.8$. It is precisely $t_1 \neq t_2$ that realizes altermagnetism; $t_1 = t_2$ recovers an ordinary Néel antiferromagnet with spin-degenerate bands.

3 Method

Diagonalize the two 2×2 spin blocks over a uniform BZ grid (coarse-first $n_k = 24 \to 48 \to 96$, SAVE-EARLY after each). Measure: (a) net magnetization; (b) lower-band spin splitting $\Delta(k) = E^{lo}_{\uparrow} - E^{lo}_{\downarrow}$ across the BZ; (c) splitting on the diagonals (node test); (d) sign along Γ–X vs Γ–Y (C4 sign flip); (e) C4 antisymmetry residual $\max|\Delta(k_x, k_y) + \Delta(k_y, k_x)|$; (f) agreement of the numeric splitting sign with the analytic d-wave $\delta(k)$.

4 Results

Quantity	$n_k = 24$	$n_k = 48$	$n_k = 96$
M_{net} / cell	0	0	0
BZ-avg splitting	6×10^{-18}	-6×10^{-18}	-6×10^{-18}
max $\lvert\Delta\rvert/t_{nn}$	0.5612	0.5612	0.5612
diagonal-node max $\lvert\Delta\rvert$	0	0	0
$\langle\Delta\rangle_{\Gamma X}$	−0.2246	−0.2246	−0.2246
$\langle\Delta\rangle_{\Gamma Y}$	+0.2246	+0.2246	+0.2246
C4 antisymmetry residual	0	0	0
d-wave sign match	100%	100%	100%

Table 1: Converged (grid-independent) at every n_k; these are symmetry-exact quantities.

5 Comparison to the paper

Claim (paper)	This work	Verdict
Vanishing net magnetization	$M_{\text{net}} = 0$ exactly	✓
Conserved spin channels	S_z good (no SOC), block-diagonal	✓
Momentum-dependent spin splitting	$\max\lvert\Delta\rvert = 0.561\, t_{nn}$	✓
d-wave symmetry / diagonal nodes	nodes to 10^{-16}	✓
Sign flip under C4	ΓX vs ΓY opposite; antisym 10^{-16}	✓
Combined spin×rotation symmetry	C4 maps $\Uparrow\!\leftrightarrow\!\Downarrow$ iso-surfaces	✓

All six qualitative-but-symmetry-exact ingredients of the headline claim reproduce.

6 Critique, gaps and scope

This is a *minimal d-wave prototype*, not a material-specific calculation. Not built: (i) g-wave/i-wave cases (the paper's real materials MnTe, CrSb are g-wave); (ii) ab-initio band structure; (iii) the superfluid-^{3}He / higher-partial-wave Pomeranchuk analogy that forms the intellectual core of the Perspective; (iv) relativistic (SOC) altermagnetic spin-splitting effects. Because the paper states no number, coverage is inherently capped: we reproduce the *mechanism and its symmetry protection*, which is the falsifiable content, but there is no quantitative benchmark to agree with. Verdict: **REPLICATED** at the mechanism/symmetry level; the final verdict is assigned by the LLM judge.